Environmental Impact Assessment of the 1970 Bhola Cyclone

Introduction

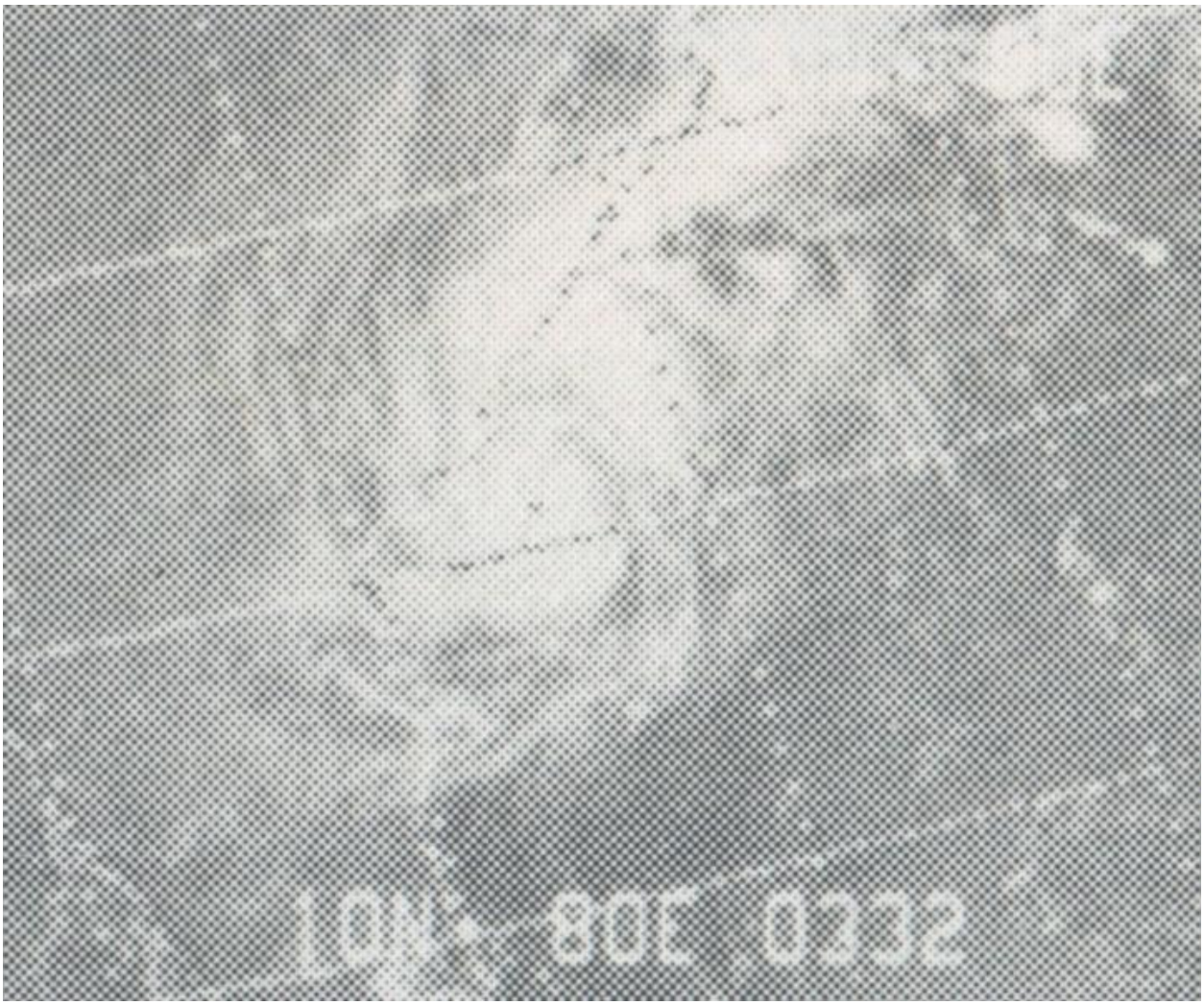

Figure: Satellite image of a tropical cyclone in the Bay of Bengal (4 Nov 1976). Such storms produce massive storm surges that inundate low-lying coasts.

Tropical Cyclone **Bhola** struck East Pakistan (now Bangladesh) on 12 Nov 1970 as an extremely severe cyclone. Its ~35-foot storm surge swept over barrier islands and flat delta plains, flooding villages and polders. Bhola killed roughly 300,000–500,000 people, making it the deadliest tropical cyclone on record. Millions of survivors were displaced and agriculture was devastated.

Although modern EIA laws did not exist in 1970, today national frameworks (e.g. the U.S. NEPA/EPA, Clean Water Act, etc.) and Bangladesh's laws (e.g. Forest Act 1927, Environment Conservation Act 1995, Water Act 2013) require systematic impact assessments. Even for historical disasters like Bhola, the **concept** of an Environmental Impact Assessment can be applied in retrospect. For example, legislation typically specifies criteria for screening and full EIA, and one could *screen* the Bhola event as a high-impact "project." The following analysis uses the EIA framework (scoping, checklists, matrices, etc.) to organize and evaluate Bhola's environmental effects and recovery needs.

EIA Methodologies

- **Screening:** Determine if an EIA is needed. Normally, laws set thresholds or criteria. Retrospectively, we "screen in" Bhola as a catastrophic event clearly warranting full analysis (by any modern EIA standard).
- **Scoping:** Identify the key environmental issues and boundaries. Scoping defines which media and receptors to study and involves setting spatial/temporal limits. Techniques include baseline surveys, checklists, matrices and network diagrams. The scoping process would involve local stakeholders (policymakers, survivors, NGOs) to highlight concerns like water safety, agriculture losses, housing, and health.
- **Checklists & Matrices:** Use structured tools to ensure no major impact is overlooked. For example, a checklist might list categories (water, soil, biota, etc.), and a Leopold-style matrix could link storm features (surge, wind, flooding) to receptors (rivers, soils, crops, human health). These tools help systematically identify, qualify and quantify impacts.
- **Cost–Benefit Analysis:** Estimate the economic value of impacts and compare to mitigation. Here one would quantify the cost of damages (lost crops, homes, fisheries, infrastructure) versus costs of protective measures. As FAO notes, EIA comparisons of alternatives should include "potential impact and cost/benefit analysis"[1]. For example, one could compare the benefit of building embankments or shelters versus the huge loss of life and property that occurred.
- **Public Participation:** Engage communities in the assessment. Modern EIA emphasizes stakeholder input at all stages. In Bhola's case, survivors' experiences could be gathered via interviews or public meetings to document unquantified effects (cultural sites lost,

traditional resource impacts) and priorities for relief. In particular, scoping should "involve community participation to some degree", ensuring that local knowledge and needs inform the analysis.

Components of the Environment – Water, Land, Soil

- **Surface Water:** Bhola's storm surge and floods completely overwhelmed ponds, rivers and shallow lakes. Seawater inundated surface-water bodies (rivers, canals, marshes), mixing salt, sewage and debris into drinking sources. Freshwater ponds and wells turned saline, endangering potable water. For example, a relief survey found unusually high salinity in some surface waters after Bhola[2]. Public health was a major concern: experts feared cholera and typhoid outbreaks due to water contamination[3] (though later surveys found no epidemic ultimately occurred[4]). Floodwaters also carried sediments and pollutants into water channels, altering water quality and aquatic habitat.

- **Groundwater:** The prolonged inundation raised groundwater salt content. Tropical storm surges are known to drive saltwater into coastal aquifers. Studies confirm that "historic tropical cyclone induced storm surges have contributed to the high groundwater salinities observed in coastal Bangladesh"[5]. After Bhola, many shallow wells and tube wells on low islands became brackish or undrinkable. Communities often relied on rainwater ponds and shallow wells; these sources were inundated by saline water. The elevated groundwater salinity would have persisted for months, forcing people to travel farther for fresh water or suffer health effects.

- **Coastal Landforms:** Bhola's surge dramatically reshaped the shoreline. NOAA reports that the 10–12 m surge "washed over many barrier islands and the flat shoreline of the Bay of Bengal". Many small islands (chars) in the Ganges–Brahmaputra delta were submerged or eroded. The cyclone scoured river channels and deposited sand and mud inland, altering island coastlines. The outlying Bhola Island itself was almost bisected by floodwaters. Although detailed geomorphic surveys from 1970 are scarce, we know villages and trees were literally wiped away on these landforms.

- **Soil Erosion:** Floodwaters and winds stripped topsoil. Bhola's surge washed salt onto polders and rice paddies, degrading soil fertility. Meanwhile, strong currents eroded away

surface soil layers. Agricultural impact was severe: one study estimates Bhola "smashed crops on around 17% of the land area and destroyed around 30% of food production" in Bhola district. This implies massive loss of arable topsoil. In addition, sediments deposited by the surge would have buried some soils in layers of sand or silt. In sum, fertile soils were salinized, eroded and in many places replaced by mud, causing long-term declines in crop yields.

Components of the Environment – Air, Noise, Biota, Socio-Economic

- **Air Quality and Noise:** During the cyclone, gale-force winds would have churned air and generated intense noise, but these effects were transient. High winds likely cleared dust and pollutants from the air; on the other hand, sea spray and aerosolized salt may have briefly increased aerosol concentrations. After Bhola, stagnant floodwaters and rotting debris could have produced foul odors and mosquito-borne pathogens, but quantitative air-pollution data are not available. Overall, apart from the storm's roar and tearing winds, long-term air pollution impacts were minor compared to water and land effects.

- **Flora and Fauna (Biota):** Bhola devastated terrestrial and coastal ecosystems. All vegetation (crops, grasses, trees) in the inundated zone was uprooted or buried in salt. Cropland and orchard trees were destroyed by wind and salt spray. Wetland plants (mangroves, marsh grasses) were submerged for days and heavily stressed or killed. Wildlife also suffered: many fish, amphibians, reptiles and mammals in flooded areas drowned or were displaced. Coastal wetlands lost bird and fish habitats. For example, modern observations show that when Bangladesh cyclones submerge the Sundarbans mangroves for days, freshwater ponds and wildlife suffer dramatic losses[6] – implying that Bhola likely caused similar biotic stress (although specific surveys from 1970 are lacking).

- **Socio-economic and Cultural:** The human toll and social disruption were enormous. Bhola killed an estimated 300,000–500,000 people, mostly the rural poor. Villages were flattened and ~3 million people lost homes. Traditional livelihoods vanished: farmers lost their land (as noted above) and fishermen lost boats and gear. In Bhola district alone, about 115,000 fishermen were reported impacted. The disaster destroyed schools, temples and

community sites along with infrastructure (roads, wells). Health facilities were overwhelmed – relief agencies feared cholera epidemics[3] and indeed saw evidence of waterborne disease (though surveys found no outbreak[4]). The cultural fabric was also affected: there are accounts that Bhola's catastrophe "re-made South Asia" by fueling political upheaval. The Pakistani government's slow response in East Pakistan led to widespread outrage, helping the Awami League win elections and ultimately sparking Bangladesh's independence struggle[7]. This illustrates how an environmental disaster had profound social and cultural consequences.

Environmental Management Plan

- **Early Warning and Shelters:** Bangladesh has since invested in early-warning systems and cyclone shelters that Bhola-era residents lacked. The Cyclone Preparedness Programme (1973) now mobilizes thousands of volunteers with sirens and radios to warn coastal villagers. During later cyclones, hundreds of shelters and evacuation routes save lives. Continual maintenance of these systems is critical. For example, in Cyclone Amphan (2020), over 2 million people evacuated to shelters thanks to Bangladesh's preparedness[8]. Such measures should be continuously improved: regular community drills, robust communication (megaphones, text alerts), and secure infrastructure ensure future storms have lower human cost.

- **Coastal Defenses and Restoration:** Building and reinforcing embankments and dikes can physically block storm surges. Complementarily, restoring natural buffers greatly reduces impact. Planting mangrove belts along coastlines absorbs wave energy and traps sediments. Bangladesh's investment in mangrove afforestation (with GEF and local efforts) is an example of this approach. Land-use planning should enforce minimum elevation for housing: allowing only salt-tolerant crops or fish ponds in the most exposed areas. Infrastructure (roads, bridges) can be designed to withstand flooding (e.g. raised on stilts). In essence, combining "hard" engineering with "soft" ecosystem-based measures builds long-term resilience.

- **Environmental Management Systems (EMS) and Audits:** At the organizational level, authorities and NGOs can adopt ISO 14001-style EMS to manage environmental risks.

ISO 14001 specifies a framework for an effective Environmental Management System[9], ensuring continual improvement. For disaster contexts, an EMS could set objectives such as minimizing pollution during relief (e.g. proper disposal of waste), conserving resources, and restoring habitats after floods. Periodic **environmental audits** – systematic, documented evaluations of performance[10] – would check that the EMS is working. For example, after a cyclone response, auditors could assess how well water and waste management met standards, or whether hazard maps were updated. Such a formal process institutionalizes lessons learned and holds agencies accountable.

- **Hazard/Risk Analysis and Monitoring:** Cyclone risk should be incorporated into planning tools. Governments can require **risk assessments** for development projects: evaluating the probability of storm surge inundation, wind damage, or flooding for any new infrastructure. High-risk zones (as shown on hazard maps) might be zoned for green space or salt-tolerant uses instead of housing. Remote sensing and real-time monitoring (satellites, tide gauges) provide data to update risk models. After Bhola, one could imagine a retrospective "risk register" listing all major cyclone hazards (surge, winds, pollution) and planned mitigations. Mandating post-disaster EIA-like reports would document impacts and improvements needed – in effect closing the loop of adaptive management.

- **Policy and International Frameworks:** Finally, policy measures and standards can reinforce these actions. Bangladesh has strengthened its disaster laws and integrated DRR into national plans since Bhola. Environmental regulations (e.g. the Water Act) should be enforced even during relief (for instance, controlling spills from flooded latrines and fuel stores). At the highest level, Bangladesh adheres to frameworks like the UN Sendai Framework for Disaster Risk Reduction, which emphasizes building back better. International support (UN, Red Cross) also promotes best practices. In sum, a combination of ISO/EMS tools[9], environmental auditing[10], risk analysis, and strong regulatory enforcement creates a post-Bhola management plan that could mitigate future cyclones and protect both people and ecosystems.

Sources: Authoritative meteorological reports, academic studies, and government sources were used. Facts on Bhola's impacts are drawn from NOAA and WMO retrospectives, historical analyses[11], and public health surveys[3][4]. EIA methods and management ideas refer to FAO and ISO guidelines[10][9][8]. Additional context comes from peer-reviewed studies on salinity and water quality[5] and Bangladesh laws and programs as cited.

[1] [10] Environmental impact assessment and environmental auditing in the pulp and paper industry

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